WORKING PAPER

The Rural Mobilization Gap in U.S. Place-Based Tax Credit

Intermediary Selection versus Market Structure in the New Markets Tax Credit

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This draft: August 2026 · Comments welcome

ABSTRACT

The U.S. New Markets Tax Credit (NMTC), enacted in 2000, gives private investors a 39% federal tax credit for routing capital through certified Community Development Entities (CDEs) into projects in low-income census tracts. Over FY2001–FY2022, the program channeled \$66.6 billion of subsidized investment into 8,024 projects worth \$120.9 billion. Each dollar of subsidized investment sat alongside \$0.82 of other capital. This is a financing multiple. Federal cost equals the 39% credit on the equity basis; using tax expenditure as the denominator gives a ratio of roughly \$2.09. Non-metropolitan tracts received 19.6% of dollars, close to the program's 20% administrative target, yet project-level leverage was lower there (median 1.07× versus 1.19× metro).

We decompose this rural mobilization penalty with project-level fixed-effects regressions using the CDFI Fund's public release. The raw estimate is a precise −0.262 rural penalty in mean leverage. Year, project-type, and state fixed effects shrink it to −0.172. Substituting CDE effects for state effects moves the coefficient to −0.047; a fully nested model with both gives −0.060. The mean regression is imprecise in either form (cluster-robust 95% CI [−0.245, +0.152]). The median-regression analog is a precise −0.001 (CDE-cluster bootstrap SE 0.009 over 500 replications), and both the extensive margin (any private mobilization) and the intensive margin are null within intermediary. Estimating the same specification across the quantile process leaves point estimates drifting negative above the median, though the intervals widen at least as fast and no upper-tail quantile is distinguishable from zero. An order-invariant Gelbach decomposition attributes −0.185 of the raw gap to CDE identity (CDE-cluster bootstrap 95% CI [−0.347, −0.023]; 86% of the explained movement), compared with −0.041 for project type. The rural leverage gap lies largely between CDEs. Re-estimating the specification inside eighteen subgroups, by intermediary size, era, region, project type and rural orientation, produces no cell with a within-CDE penalty that survives a multiple-comparison correction.

A cross-CDE excess-mass test at the 20% non-metropolitan target finds no excess mass in cumulative deployment. This holds for deal-count and dollar shares, over the full period and after the proportionality rule entered the code in 2006 (dollar-share excess mass $\hat{B} = 0.0009$ after 2007; cluster-bootstrap 95% CI [−0.013, +0.016]). Compliance may operate through allocation agreements, which the deployment data do not record. For policy, the relevant levers concern intermediary selection and capacity; market-structure redesign addresses a different margin.

JEL H25 · H81 · R51 · G28

Keywords New Markets Tax Credit · place-based policy · blended finance · mobilization ratio · Community Development Entities · rural finance

1 Introduction

The blended-finance literature has long debated which structures best mobilize private capital alongside public concessional finance [Convergence Finance, 2024, Attridge and Engen, 2019, Lankes, 2021]. Its recurring empirical problem is measurement. The "mobilization ratio," private dollars moved per public dollar committed, is rarely observed at the project level. Most evidence comes from survey reports or coarse program aggregates.

The U.S. New Markets Tax Credit offers a setting where the financing structure behind that ratio becomes visible at the deal level. For every project, the CDFI Fund's public release records both the subsidized investment that the intermediary places (the Qualified Low-Income Community Investment, or QLICI) and total project cost. We use their ratio, the amount of project capital accompanying each dollar of subsidized investment. Section 3 defines the measure more fully. The government's cost is the credit claimed against the equity basis, so fiscal leverage uses a different denominator. We report both denominators when magnitudes matter. The panel follows one instrument in one country for twenty-two years and roughly 8,000 projects, with the financing structure observed directly.

Projects in non-metropolitan tracts carry less capital beyond the subsidized investment than projects in metropolitan ones. The question is where that gap originates. It may reflect *market structure*, with rural projects fundamentally less leverageable. It may instead reflect *intermediary selection*: the Community Development Entities active in rural tracts may mobilize less capital everywhere they operate. The policy prescriptions differ. Market structure points toward redesigning the credit; selection points toward reallocating it.

The estimates point to composition. The raw rural penalty in project-level leverage is −0.262 and precisely estimated. Origination-year, project-type, and state fixed effects move it to −0.172. CDE fixed effects compare rural and urban deals *made by the same intermediary*, using the 163 of 343 CDEs that work on both sides of the line. With those effects, the point estimate falls to −0.047; a fully nested model retaining state effects gives −0.060. The mean regression remains imprecise (cluster-robust 95% CI [−0.245, +0.152]) and can reject only within-CDE penalties larger than 0.21. The other estimates are tighter. The median within-CDE penalty is −0.001 (SE 0.008). Within a CDE, rural deals are equally likely to draw capital beyond the subsidized investment and draw no less when they do. The spatial difference comes from which intermediaries originate the deals.

The NMTC evaluation literature has examined program targeting and neighborhood outcomes extensively [Gurley-Calvez et al., 2009, Freedman, 2012, Harger and Ross, 2016, Freedman and Kuhns, 2018, Theodos et al., 2020, 2022, Corinth et al., 2024], with program-evaluation foundations in Abravanel et al. [2013]; the place-based policy literature supplies the welfare frame [Glaeser and Gottlieb, 2008, Kline and Moretti, 2014, Neumark and Simpson, 2015, Diamond and McQuade, 2019]. Mobilization has received much less attention in either literature. We have found no prior econometric study that treats project-level financing structure as the main NMTC outcome. Doing so brings the mobilization framework to work otherwise centered on neighborhood employment, demographic change, and price effects.

We then use CDE fixed effects to separate the within-CDE rural penalty from the between-CDE selection component. Descriptive work by Theodos et al. [2021] motivates this decomposition without estimating it. I also apply the excess-mass logic of Chetty et al. [2011] and Kleven [2016] to the cross-CDE distribution of non-metro deployment shares around the program's 20% target. We have not found this test in the published literature. There is no excess mass at the 20% line in deal counts or dollars, either over the full period or after the proportionality rule entered the code. One explanation is that compliance operates through allocation agreements absent from the public release.

Section 2 describes the program and its CDE intermediary layer. Section 3 covers the public release and construction of the leverage outcome. Section 4 sets out the fixed-effects design and its limits, and Section 5 reports the decomposition, heterogeneity, robustness, and bunching diagnostic. Section 6 gives the policy reading; Section 7 concludes. A causal companion design, using a regression discontinuity at the low-income-community eligibility threshold, requires a tract-level ACS merge and remains for follow-on work.

2 Institutional background

2.1 *Program mechanics*

The NMTC, enacted in the Community Renewal Tax Relief Act of 2000, awards investors a 39% federal tax credit, claimed over seven years, on Qualified Equity Investments (QEIs) made into certified Community Development Entities. CDEs in turn deploy that capital as Qualified Low-Income Community Investments into Qualified Active Low-Income Community Businesses (QALICBs) located in eligible census tracts. The public release distinguishes four QALICB types: real-estate projects (RE), non-real-estate operating businesses (NRE), special-purpose entities (SPE), and CDE-to-CDE loans. These mechanics put the CDE at the center of the program. It holds the allocation and originates the deals, then structures the capital stack around the subsidized tranche. Figure 1 shows the sequence. Capital moves from the investor through the CDE to the project, while the tax credit returns to the investor.

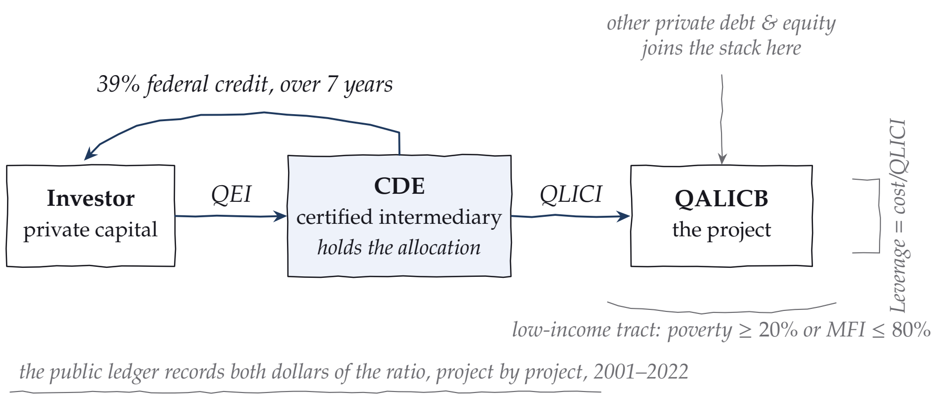

Figure 1. The program, as a working sketch. Private capital enters as a Qualified Equity Investment. The intermediary deploys it as a QLICI into a project in an eligible tract, and the 39% credit returns to the investor over seven years. The public ledger records both inputs to the leverage ratio for every project.

2.2 *Eligibility and the 20% non-metro minimum*

Eligible low-income community (LIC) tracts are defined by a poverty rate of at least 20% or median family income at or below 80% of the area benchmark, with a targeted-populations override. Relevant here, a later amendment directs the Treasury to write rules ensuring that non-metropolitan counties receive a proportional allocation of qualified equity investments. That instruction is IRC §45D(i)(6), added by the Tax Relief and Health Care Act of 2006 (P.L. 109–432, §102(b)), so it postdates the program's creation in the Community Renewal Tax Relief Act of 2000 (P.L. 106–554, §121), leaving the earliest sample years outside its scope. The statute names no percentage. The familiar 20% figure comes from the CDFI Fund's administration of the requirement through allocation applications and agreements. We therefore use "administrative target" throughout.[1] Aggregate deployment is close to that line: non-metro tracts received 19.6% of QLICI dollars over the sample period.

2.3 *The intermediary layer*

The public release names 345 distinct CDEs in the transaction ledger (343 at the project level); 310 executed five or more transactions. The twenty largest account for 23.9% of QLICI dollars, so deployment is moderately concentrated. Specialization is extreme: among CDEs with at least five

[1]The effective dates matter for the analysis; Section 5.6 reports the mandate results for the full period and for origination years from 2007 onward.

transactions, cumulative non-metro shares span the full range from 0% to 100%, with 39% of such CDEs never deploying non-metro and 6.5% deploying there at least four times out of five. This dispersion motivates the decomposition. An aggregate rural penalty can arise when rural deals flow through intermediaries that mobilize less *everywhere*, even if their rural and urban projects have similar leverage. The public release identifies CDEs by name. A later extension will classify their institutional form (bank subsidiary, nonprofit CDFI, for-profit specialist).

3 Data

The analysis uses the CDFI Fund's NMTC Public Data Release covering FY2003–FY2022 award years (released June 2024), which reports 19,907 QLICI transactions across 8,024 projects with origination years 2001–2022. Transactions are aggregated to the project level; the project is the unit at which total cost, and hence leverage, is defined. Cleaning steps, field definitions, and file hashes are documented in the public repository's data dictionary and provenance files, and the full pipeline from the raw workbook to every table in this paper runs from a single script sequence.

The outcome is project-level leverage,

$$\text{Leverage}_i = \frac{\text{ProjectCost}_i}{\text{QLICI}_i},$$

which we refer to as leverage throughout. Leverage is bounded below at 1 by construction, and the mass at 1 collects projects whose recorded cost equals the subsidized investment alone.

The ratio measures financing structure. The QLICI is the investment placed by the intermediary; federal cost comes through the 39% credit claimed against the qualified equity investment that funds it. That credit is spread across seven years. Under the substantially-all requirement linking the two magnitudes, a dollar of QLICI corresponds to roughly thirty-nine cents of tax expenditure. The aggregates show the difference. $120.9 billion of project cost against $66.6 billion of QLICI leaves $54.3 billion of other capital, or $0.82 for every dollar of subsidized investment and about $2.09 for every dollar of implied tax expenditure. A constant change of denominator leaves the rural coefficient's sign, its relative magnitude, and every test below unchanged. The baseline outcome is winsorized to $[1, 20]$ to discipline a long right tail of large capital-stack deals; Section 5.4 shows the headline result is insensitive to the cap, including its removal.

Table 1. NMTC deployment by metro status, FY2001–FY2022.

	Metro	Non-metro
Projects	6,463	1,561
QLICI deployed ($M)	53,566	13,045
Total project cost ($M)	97,151	23,751
Mean leverage	1.99	1.73
Median leverage	1.19	1.07

Table 1 reports descriptives. Non-metro projects are 19.5% of the sample, again close to the administrative target, and lag metro projects in both mean (1.73 versus 1.99) and median (1.07 versus 1.19) leverage. Figure 3 shows the shape of the gap: non-metro mass piles up near the 1.0 floor while the metro distribution carries a fatter right shoulder.

4 Empirical strategy

4.1 *Layered fixed-effects decomposition*

Let R_i indicate a non-metro project. We estimate the sequence

$$L_i \;=\; \alpha + \beta R_i + \delta_{t(i)} + \eta_{q(i)} + \mu_{s(i)} + \gamma_{c(i)} + \varepsilon_i,$$

adding origination-year (δ), QALICB-type (η), state (μ), and CDE (γ) fixed effects in turn. The workhorse CDE specification substitutes intermediary effects for the state effects in the preceding column, since intermediary identity already absorbs much of the same variation. Column 7 of Table 2 carries both sets of effects and provides the strictly nested comparison. With $\gamma_{c(i)}$ included, $\hat{\beta}$ comes from rural–urban comparisons *within* an intermediary. It measures the within-CDE rural penalty, or the market-structure component. The change in $\hat{\beta}$ after adding CDE effects measures the between-CDE selection component. We cluster standard errors in these specifications at the CDE level, where both the fixed effect and the economic decision reside.

4.2 *Interpretation and limits*

This is a selection-on-observables decomposition. CDEs do not allocate deals randomly across rural and urban tracts, so we give $\hat{\beta}$ no LATE inter-

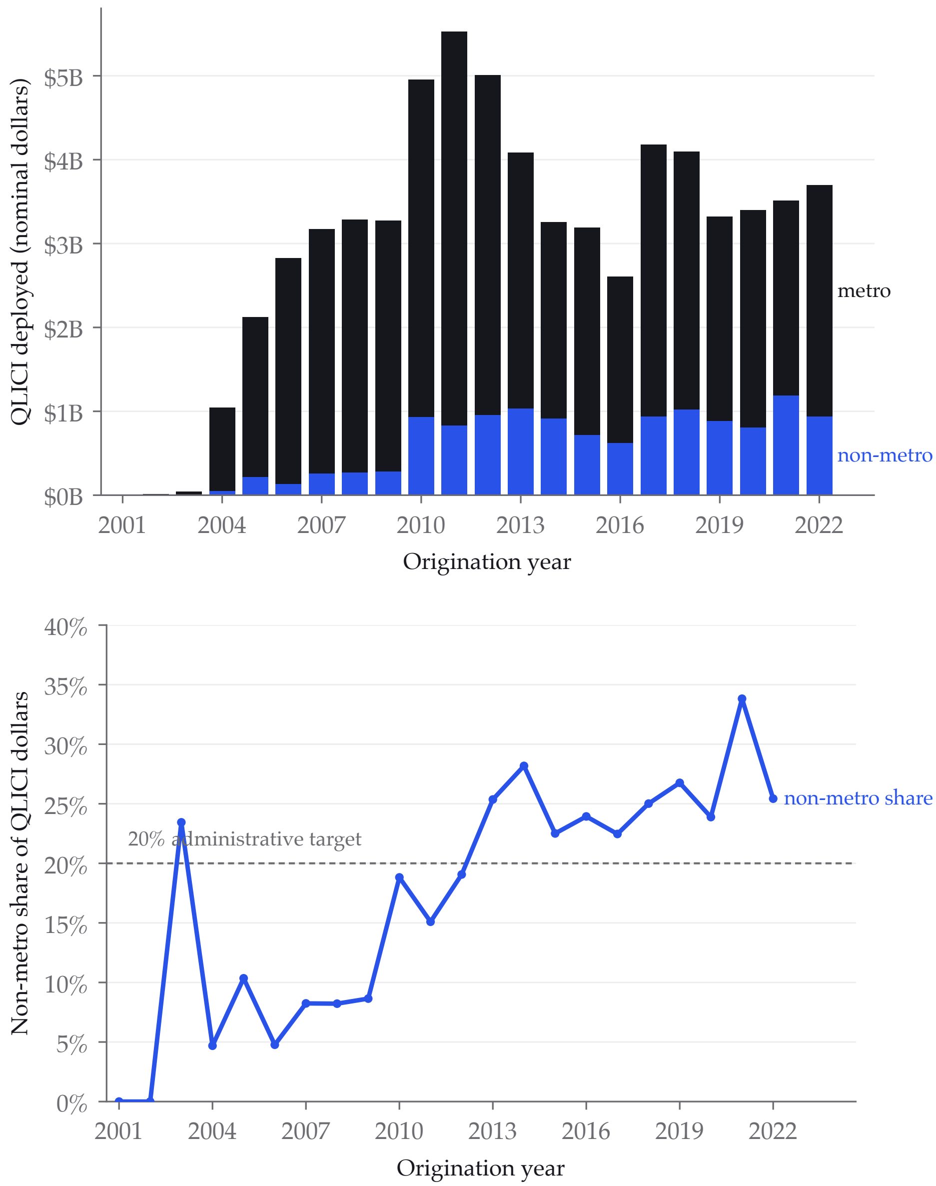

Figure 2. Deployment over time. Left: annual QLICI dollars, metro and non-metro. Right: the non-metro dollar share against the 20% administrative target.

pretation. The design establishes whether the aggregate rural penalty survives comparison within the intermediary. That comparison holds fixed its deal-structuring capacity, underwriting practice, and investor relationships. A fuzzy RDD at the LIC eligibility threshold, estimated separately for metro and non-metro samples, requires tract-level ACS covariates and remains for the follow-on paper.

The between-CDE component is an observed composition fact. Call-

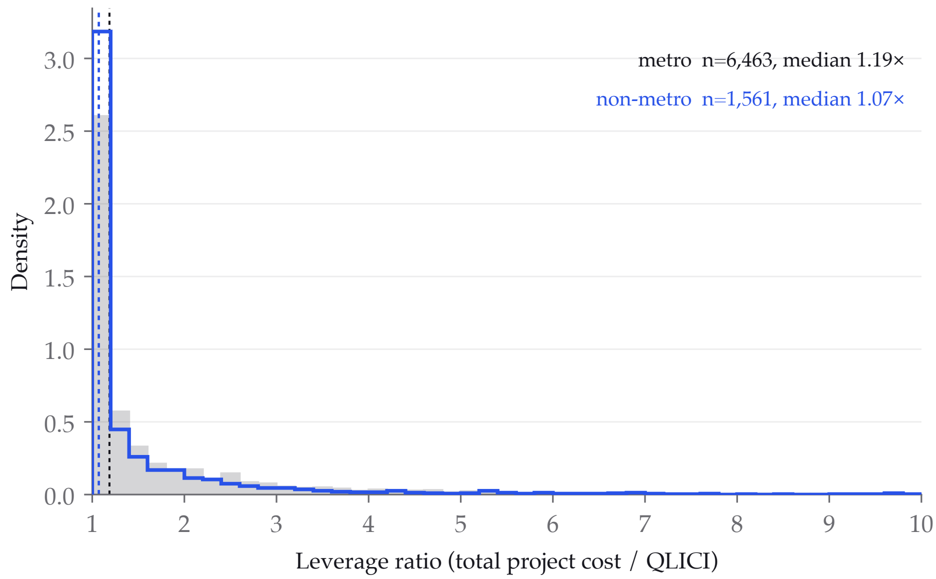

Figure 3. Project-level leverage distributions by metro status. Non-metro mass concentrates at the 1.0× floor (zero private mobilization); the metro distribution has the fatter right shoulder.

ing it "selection" adds an interpretation. Rural market conditions could cause low-leverage intermediaries to specialize in rural deployment, placing part of the between-CDE component in market structure operating through their business models. The within-CDE null is directly identified.

4.3 *Excess mass at the 20% target*

For each CDE j, define its cumulative non-metro deployment share s_j, computed both as a count share, $n_{j,\text{nonmetro}}/n_{j,\text{total}}$, and as a dollar share of QLICI, since the requirement is written in terms of investment. If the 20% target binds at the deployment margin, optimizing CDEs should pile up at $s_j = 0.20$ and produce excess mass in the sense of Chetty et al. [2011] and Kleven [2016]. The test compares empirical mass in a ±2.5 percentage-point window around 20% against a degree-3 polynomial counterfactual fitted outside the window, over the 310 CDEs with at least five transactions, with a CDE-level bootstrap for inference. The public data reveal no discontinuity in a CDE's objective or budget set at the threshold. Any consequence of falling short operates through future allocation rounds that the release does not observe. Because the proportionality instruction dates from 2006, we report the test for the full period and again for origination years from 2007 onward.

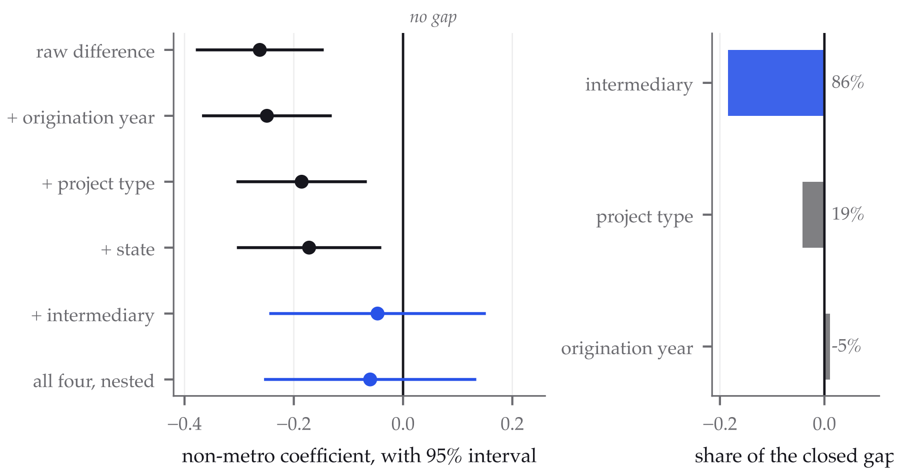

Figure 4. The rural coefficient as fixed effects enter. Left, the non-metro estimate with its 95% interval under each specification; the interval crosses zero only when intermediary identity enters, and stays there in the strictly nested model. Right, the order-invariant decomposition of the movement between the raw and full specifications. Values are read from the same pipeline outputs that produce Tables 2 and 3.

5 Results

5.1 The decomposition

Figure 4 puts the whole argument on one axis before the tables spell it out. Each interval is the non-metro coefficient under a successively richer specification, and the collapse happens at one step.

Table 2 reports the main estimates. The raw rural gap (column 1) is −0.262 with a heteroskedasticity-robust standard error of 0.060: non-metro projects carry about a quarter of a dollar less additional capital per dollar of subsidized investment, and the difference is precise. Origination-year effects barely move it (column 2, −0.249), leaving little scope for a vintage explanation. QALICB-type effects (column 3) shrink it to −0.186. Rural deals skew toward operating-business types with lower leverage, and this composition explains roughly a quarter of the raw gap. State effects (column 4) move it modestly further to −0.172.

Introducing CDE fixed effects in column 5 collapses the rural coefficient to −0.047 with a CDE-clustered standard error of 0.101, roughly a fifth of the raw gap in point-estimate terms. The interval, [−0.245, +0.152], is wide, and Section 5.2 quantifies how much the mean regression can and

Table 2. The rural leverage gap under layered fixed effects. Outcome: project leverage, winsorized $[1, 20]$. Columns (1)–(4) report HC1 standard errors; columns (5) and (7) cluster at the CDE level; column (6) reports the quantile-regression asymptotic standard error, which is not clustered. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) OLS	(6) Median	(7) OLS
Non-metro ($\hat{\beta}$)	-0.262***	-0.249***	-0.186***	-0.172**	-0.047	-0.001	-0.060
	(0.060)	(0.061)	(0.061)	(0.067)	(0.101)	(0.008)	(0.099)
Year FE	–	✓	✓	✓	✓	✓	✓
QALICB-type FE	–	–	✓	✓	✓	✓	✓
State FE	–	–	–	✓	–	–	✓
CDE FE	–	–	–	–	✓	✓	✓
R^2	0.002	0.015	0.022	0.039	0.131	–	0.141
N	8,024	8,024	8,024	8,024	8,024	8,024	8,024

cannot rule out. The median specification (column 6) is precise, placing the within-CDE rural penalty at -0.001 (SE 0.008). Because column 5 substitutes CDE effects for the state effects of column 4, column 7 reports the strictly nested model carrying year, type, state, and CDE effects together; it returns -0.060 (SE 0.099, $p = 0.54$), so no part of the reading depends on that substitution.

Because sequential addition of fixed-effect blocks is order-dependent, Table 3 reports the Gelbach [2016] decomposition with the full model as reference, which is invariant to listing order. Of the -0.216 movement from the raw to the full specification, CDE identity contributes -0.185 (86% of the movement; CDE-cluster bootstrap 95% CI $[-0.347, -0.023]$), QALICB type -0.041, and origination year a small offsetting $+0.010$. The share version of this statement, 82% of the raw gap, carries a wide bootstrap interval ($[25\%, 255\%]$) because it is a ratio of estimates. The level estimate is more informative. The CDE composition component is large, negative, and statistically distinct from zero. The within-CDE component is statistically indistinguishable from zero.

5.2 Precision, power, and margins

The mean and median estimates differ in precision by an order of magnitude. For the mean specification, a one-sided 5% test can reject within-CDE rural penalties larger than 0.213, nearly the size of the raw gap. The mean regression alone therefore gives weak evidence for a zero; the outcome's

Table 3. Gelbach (order-invariant) decomposition of the raw rural gap, full-model reference. Contributions sum to $\hat{\beta}_{M0} - \hat{\beta}_{\text{full}}$ by identity (verified to 10^{-14} in the replication code).

	Contribution	Share of movement
Origination year	+0.010	-4.8%
QALICB type	-0.041	19.2%
CDE identity	-0.184	85.6%
Total ($\hat{\beta}_{M0} - \hat{\beta}_{\text{full}}$)	-0.216	100%
CDE component bootstrap 95% CI: [-0.347, -0.023] (499 CDE-cluster reps)		

long right tail, which inflates mean-regression variance, is the proximate reason. The median specification can reject penalties larger than 0.016. The median estimate and the margins carry the null; we report the mean for comparability.

Because the median estimate carries so much of the argument, we subject its precision to three further tests, reported in Table 4. A CDE-cluster pairs bootstrap over 500 replications, refitting the full median regression on each draw and treating an intermediary drawn twice as two intermediaries with their own fixed effects, returns a standard error of 0.0093 against the asymptotic 0.0076. Clustering therefore inflates the median standard error by about a quarter, which moves the rejectable penalty from 0.013 to 0.016 and leaves the substantive claim intact. Randomization inference, permuting the rural label inside each intermediary so that every book composition is held fixed, returns a two-sided p of 0.085 against the sharp null of no within-CDE rural effect.

One caution belongs with these numbers. The outcome carries a 26.9% point mass at exactly 1.0, the value recorded when a project mobilizes nothing beyond the subsidized investment, and the unconditional median of 1.159 sits on the shoulder of that mass. Quantile-regression asymptotic standard errors are sparsity estimates that presume a positive continuous conditional density at the estimated quantile, and this configuration violates that presumption. The symptom is visible in the randomization distribution, where 91% of permutations return a rural coefficient pinned to within 10^{-7} of zero instead of varying smoothly, which is the signature of a degenerate vertex solution. We therefore report the bootstrap standard error, which makes no density assumption, and we note that it and the asymptotic one happen to agree closely here. Readers should treat the me-

Table 4. Inference for the median specification. The bootstrap is a CDE-cluster pairs bootstrap over 500 replications; randomization inference permutes the rural label within each intermediary over 400 draws. The paired test uses no regression, comparing each dual-market intermediary's rural median with its own urban median.

	Value
Point estimate, within-CDE rural coefficient	−0.0006
Asymptotic standard error (quantile regression)	0.0076
CDE-cluster bootstrap standard error (500 reps)	0.0093
Inflation from clustering	1.23×
Bootstrap percentile 95% interval	[−0.032, −0.000]
Rejectable penalty, asymptotic	0.013
Rejectable penalty, clustered	0.016
Randomization inference, two-sided p (400 draws)	0.085
Permutations pinned to a vertex	91%
Paired within-CDE median gap (104 intermediaries)	−0.018
Wilcoxon signed-rank p on the paired gaps	0.140

dian specification as well identified in its location and poorly identified in its curvature.

Figure 5 carries this further by estimating the same specification across the quantile process. The coefficient sits at zero through the lower half of the distribution, then drifts negative above the median, reaching −0.129 at the 0.90 quantile and −0.200 at 0.95. Read alone, that gradient suggests a rural penalty concentrated among the deals that mobilize the most private capital. The bootstrap does not sustain the reading. Intervals widen at least as fast as the point estimates fall, giving cluster-bootstrap standard errors of 0.125 and 0.215 at those two quantiles, so no point in the upper tail is distinguishable from zero. We report the gradient because its monotonicity is suggestive and because a design with more power in the upper tail is a natural next test, and we decline to interpret it as a result.

Figure 6 shows the identifying variation. Among the 104 dual-market intermediaries with at least three projects on each side of the rural line, each vertical spine connects a CDE's urban-book mean leverage to its rural-book mean. The spines climb a staircase from 1.0× to 4.0× across intermediaries, while the within-spine gaps distribute tightly around zero (median −0.12, interquartile range [−0.46, +0.36]). The range of levels across intermediaries is several times wider than the spread of gaps within them.

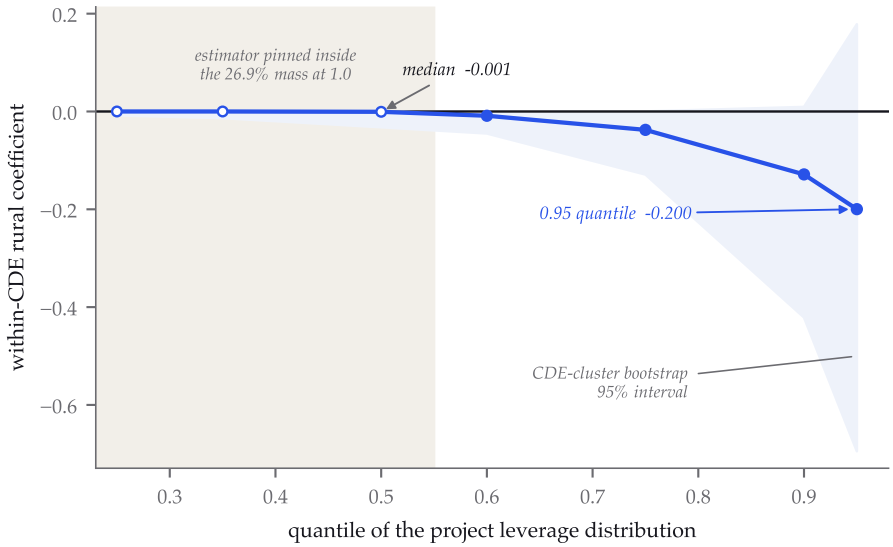

Figure 5. The within-CDE rural coefficient across the quantile process, with CDE-cluster bootstrap 95% intervals. Open markers and the shaded region mark quantiles where the estimator is pinned inside the 26.9% point mass at 1.0 and its standard error is not interpretable; the boundary is read from the share of bootstrap draws stuck at the point estimate rather than assumed. The point estimates drift negative above the median, and the intervals widen at least as quickly, so the upper tail is not distinguishable from zero.

The margins address a second way a mean could mislead: measured at a threshold of 1.001, 27.4% of projects record no capital beyond the subsidized investment, so a within-CDE penalty could in principle hide in the probability of drawing any additional capital at all. It does not. Within CDE, rural deals are 1.8 percentage points less likely to clear the floor (SE 1.8pp, $p = 0.34$), and among deals that mobilize, the rural coefficient is -0.039 ($p = 0.77$). Both margins are stable when the floor is moved to 1.05. Identification for all within-CDE results comes from the 163 of 343 CDEs that deploy on both sides of the rural line; these switchers originate 94% of all rural projects.

5.3 *Heterogeneity by project type*

Interacting the rural indicator with QALICB type (with CDE and year fixed effects retained) yields uniformly imprecise interactions: the rural main effect is 0.091 (SE 0.621), and no type-specific rural penalty is distinguishable from zero. No single project type concentrates or masks the within-CDE

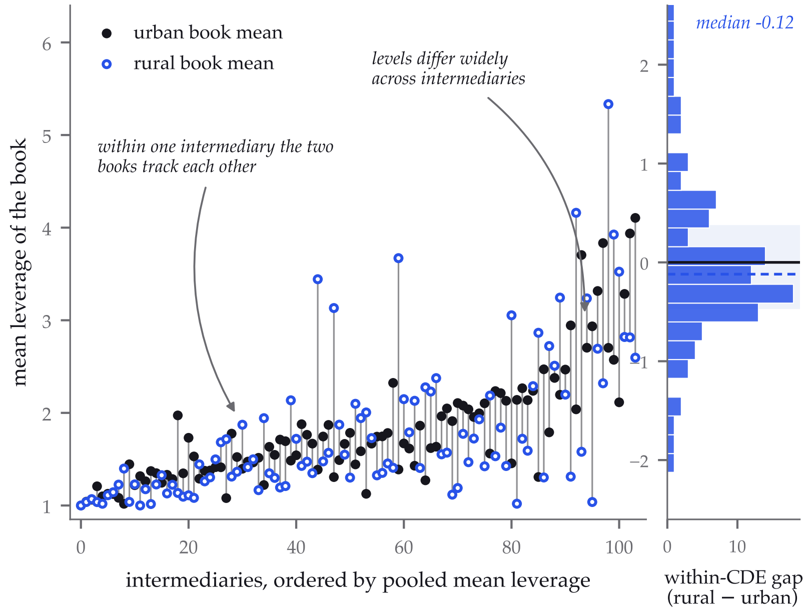

Figure 6. The switchboard. Every intermediary with at least three urban and three rural projects appears as a spine from its urban-book mean (filled) to its rural-book mean (open), ordered by pooled mean leverage; the marginal panel gives the distribution of within-CDE gaps against zero. The figure's sample rule and every displayed statistic are recorded in a machine-readable sidecar alongside the figure in the repository.

null. The point estimates weakly echo the descriptive pattern, with the most negative estimate for real estate, where private debt-stacking is most natural. The data cannot reject homogeneity.

5.4 *Robustness*

Table 5 varies the outcome transformation, winsorization cap, sample period, project sample, and clustering scheme. Unwinsorizing the outcome makes the naive gap *larger* (−0.338, $p = 0.01$), while the within-CDE estimate remains null (−0.216, $p = 0.48$). Winsorization does not produce the main result. A log outcome puts the within-CDE penalty at −3.2% ($p = 0.18$). Tightening or loosening the winsorization cap ($[1, 10]$, $[1, 50]$), splitting the sample at 2010, and restricting to single-CDE projects all leave the within-CDE rural coefficient statistically indistinguishable from zero. Restricting to the fifty largest CDEs, where the fixed effects are estimated most precisely, gives −0.100 (SE 0.148); two-way clustering by CDE and census tract moves the standard error from 0.101 to 0.105.

Table 5. Robustness of the within-CDE rural coefficient. All rows except the first include year, QALICB-type, and CDE fixed effects with CDE-clustered standard errors. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

	$\hat{\beta}$	(SE)	p	N
Unwinsorized outcome, no FE	-0.338**	(0.136)	0.01	8,024
Unwinsorized outcome	-0.215	(0.304)	0.48	8,024
Log outcome	-0.032	(0.024)	0.18	8,024
Winsorized at [1, 10]	-0.066	(0.073)	0.37	8,024
Winsorized at [1, 50]	-0.047	(0.138)	0.73	8,024
Origination 2001–2009	-0.044	(0.147)	0.77	2,488
Origination 2010–2022	-0.039	(0.107)	0.71	5,536
Single-CDE projects only	-0.088	(0.139)	0.53	6,366
Top-50 CDEs only	-0.100	(0.148)	0.50	4,229
Two-way cluster (CDE × tract)	-0.047	(0.105)	0.66	8,024
Bunching excess mass $\hat{B}$	-0.0006	95% CI [-0.0136, 0.0135], 999 CDE-bootstrap reps		

5.5 *Where the residual lives*

An average null can conceal a subpopulation, so the workhorse specification was re-estimated inside eighteen subgroups: quartiles of intermediary size, three origination eras, the four census regions, the four project types, and three bands of the intermediary's own rural orientation. One cell, the CDE-to-CDE project type, holds too few deals to identify a within-CDE comparison and is reported rather than dropped. Table 6 gives the rest.

Two of the seventeen estimated cells reach conventional significance without correction, real-estate projects at -0.395 ($p = 0.043$) and the third quartile of intermediary size at $+0.349$ ($p = 0.048$), and they point in opposite directions. Across seventeen tests at the 5% level, chance alone produces roughly one such rejection. Neither survives a Benjamini-Hochberg correction, and the smallest p-value in the entire scan is 0.043. The within-CDE null is not an average masking a subpopulation: no subgroup the data can resolve shows a rural penalty distinguishable from zero once the scanning is accounted for.

The real-estate estimate is the one worth keeping in view. It is the largest negative coefficient in the scan, it matches the descriptive pattern in the QALICB-type interaction, and real estate is where private debt-stacking is most natural, so a genuine within-intermediary rural penalty is most plausible there. The data cannot separate that reading from chance, and a design with more power in that cell would be the natural next test.

Table 6. The within-CDE rural coefficient inside subgroups. Each cell repeats the workhorse specification with year, QALICB-type and CDE fixed effects and CDE-clustered standard errors. The final column reports whether the cell survives a Benjamini-Hochberg correction at the 5% level across all seventeen estimated cells.

	$\hat{\beta}$	(SE)	p	N	switchers	BH 5%
CDE size quartile 1 (0<n≤4)	-0.838	(0.761)	0.27	203	10	no
CDE size quartile 2 (4<n≤13)	-0.179	(0.198)	0.36	670	32	no
CDE size quartile 3 (13<n≤30)	+0.349	(0.177)	0.05	1,721	53	no
CDE size quartile 4 (30<n≤inf)	-0.130	(0.122)	0.29	5,430	68	no
origination 2001-2009	-0.044	(0.147)	0.77	2,488	57	no
origination 2010-2015	+0.102	(0.173)	0.56	2,506	103	no
origination 2016-2022	-0.191	(0.128)	0.14	3,030	107	no
region Northeast	+0.090	(0.261)	0.73	1,189	26	no
region Midwest	-0.162	(0.150)	0.28	2,827	60	no
region South	+0.105	(0.141)	0.46	2,432	103	no
region West	-0.322	(0.214)	0.13	1,538	48	no
QALICB type CDE	not identified			91	4	
QALICB type NRE	+0.043	(0.153)	0.78	3,775	123	no
QALICB type RE	-0.395	(0.195)	0.04	2,862	87	no
QALICB type SPE	-0.075	(0.142)	0.60	1,296	66	no
CDE rural share ≤20%	-0.066	(0.157)	0.68	3,481	76	no
CDE rural share 20-50%	-0.023	(0.196)	0.91	1,651	57	no
CDE rural share >50%	-0.132	(0.118)	0.26	969	30	no

5.6 *No excess mass at the 20% target*

Figure 7 plots the cross-CDE distribution of cumulative non-metro shares against the polynomial counterfactual. Excess mass in the window around the 20% administrative target is $\hat{B} = -0.0006$, which is *negative* and -2.2% of the counterfactual mass, with a CDE-level bootstrap 95% confidence interval of $[-0.014, +0.013]$; 47% of bootstrap draws are positive. There is no evidence that CDEs manage their cumulative deployment to the mandate line.

We repeat the test using QLICI dollars, which matches how the requirement is written, gives excess mass of +0.005 with a bootstrap interval of $[-0.008, +0.020]$. Restricting to origination years from 2007 onward, after the proportionality instruction enters the code, gives -0.003 in counts and +0.001 in dollars, with intervals of $[-0.016, +0.011]$ and $[-0.013, +0.016]$ across 291 intermediaries.

The mandate may bind at the allocation-award level through commitments in CDE allocation agreements. The public release does not report

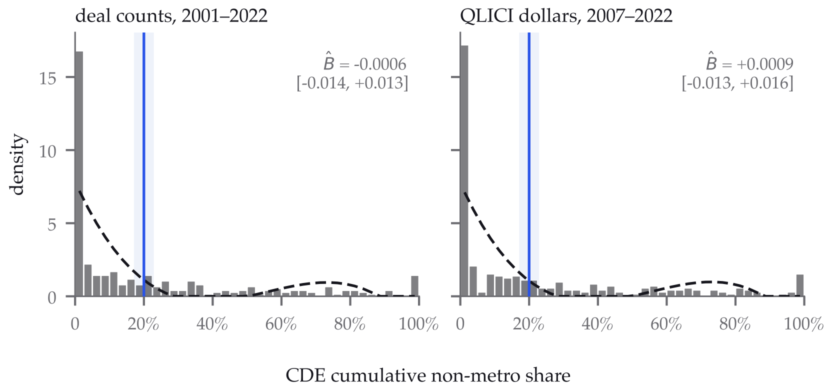

Figure 7. No excess mass at the 20% target. Cross-CDE distributions of cumulative non-metro share, over CDEs with at least five transactions, with a degree-3 polynomial counterfactual fitted outside the shaded ±2.5 percentage-point window. Left, deal counts across the full period (n = 310). Right, QLICI dollars from 2007 onward (n = 291), which matches both the units the requirement is written in and the years it governs. Each panel reports its excess mass with a CDE-cluster bootstrap interval; both contain zero. The distribution is close to bimodal, leaving the target in a low-density valley between urban and rural specialists.

those commitments, and realized cumulative deployment could fall elsewhere. The cross-CDE distribution around 20% also is close to bimodal. Urban specialists sit near zero, while rural specialists sit far above 20%, leaving the mandate line in a low-density valley.

6 Discussion

6.1 *Scope of the results*

The within-CDE rural penalty is precisely zero at the median. Both margins are null, and the mean estimate is small but imprecise. The real aggregate rural penalty lies in between-CDE composition. This does not make rural and urban credit markets equivalent. It means that NMTC deployment data detect no difference once the intermediary is held fixed.

The mechanism behind composition remains open. Intermediary selection is a natural reading of the between-CDE component. Rural market conditions could also lead low-leverage intermediaries to specialize in rural deployment, placing part of that component on the market-structure side of the ledger. Separating these accounts requires variation in CDE–market assignment that the public release lacks. The estimates also average over the fixed-effects distribution; they do not set a ceiling on what any particular CDE could mobilize in rural tracts.

6.2 *Policy reading*

Under the existing program design, the marginal rural deal does not appear constrained by market structure: a given intermediary's mobilization does not measurably fall when it goes rural. The aggregate gap is therefore amenable to *intermediary-allocation* levers. Allocations could shift toward CDEs with demonstrated high-leverage capacity, independent of their current rural orientation, without redesigning the credit itself.

The 20% mandate operates on cumulative deployment share, where the data show no bunching. If the policy goal is rural mobilization instead of rural deal counts, that is the wrong margin.

6.3 *Research program*

The next step is a multi-program U.S. comparison covering NMTC, LIHTC, and Opportunity Zones, followed by an international extension to development-bank project finance. The same mobilization framework applies in each setting. More immediately, the LIC-eligibility regression discontinuity requires the ACS tract merge.

7 Conclusion

Directly observed project-level leverage places the NMTC's aggregate rural mobilization penalty between intermediaries. The order-invariant decomposition assigns −0.185 of the raw −0.262 gap to CDE identity. The within-CDE penalty is precisely zero at the median and null at both margins. Its mean estimate is imprecise and roughly a fifth of the raw gap; a strictly nested model with state and intermediary effects returns −0.060. No subgroup we can resolve, whether by intermediary size, era, region, project type or rural orientation, shows a within-CDE penalty that survives a correction for scanning. The cross-CDE distribution of non-metro shares shows no bunching at the 20% administrative target. Intermediary composition may itself respond to rural market conditions, so its mechanism remains unsettled. For this program, the evidence points to allocation levers operating through the CDE layer.

Author contributions

Ian Helfrich assembled the project-level panel from the CDFI Fund public release, specified and estimated the decomposition, wrote the replication pipeline, and drafted the manuscript. Katia Antunes developed the blended-finance framing that motivated the study, conducted the literature review that situates it, and contributed to the interpretation of the results. Elizaveta Gonchar contributed to the empirical design and to the revision of the analysis and manuscript. All authors reviewed and approved the submitted draft.

A Reproducibility

The public repository regenerates every number in this draft. `scripts/describe_nmtc.py` builds the analysis files from the raw CDFI workbook, with hashes recorded in `PROVENANCE.md`. `scripts/run_regressions.py` produces Table 2 and the bunching statistics; `scripts/run_robustness.py` produces Table 5. `scripts/make_paper_tables.py` writes the table fragments, and `scripts/make_paper_figures.py` renders the figures included here. The repository also serves a browser-based viewer with the full project map and specification explorer.

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